The algebra behind the oracle

Companion note: sparse + low-rank structure in the H_∞ value, its gradients, and the spectral abscissa

C. Leclercq E. Clément

September 7, 2026

[This is the consolidated edition of the note: four rounds of review questions (August 31 – September 3) have been folded into the exposition, so nothing here is answered in the margin — everything is introduced in order, and the decisions the review produced are collected in §12. The September 6 edition adds §10, a projection-based reduced model for the inner loop, GL-validated and cavity-untested, with the matching to-do items. The September 7 edition credits the optimise-and-enrich loop to Aliyev, Benner, Mengi and Voigt (2019), adds the two mechanisms of Mengi (2022, 2023) for the watch list and the margin constraint, and completes the citation of the disk certificate (Schwerdtner, Mengi and Voigt 2020); these passages are set in this colour.]

The optimisation note treated the oracle as a black box. This note opens the box. The message can be stated up front: *every* quantity the optimiser consumes — the gain $\sigma_1(\omega)$, its gradient with respect to *all* design parameters at once, the location and polish of peaks, the certified global maximum, the stability margin, its certificate, and its gradient — reduces to three primitive operations:

(P1) one sparse LU factorisation, per complex point s, of the inverse of the *plant resolvent* $R(s) = (sM - A)^{-1}$ — i.e. of the sparse matrix $R(s)^{-1} = sM - A$ (the resolvent itself is never formed; "applying R" means triangular solves);
(P2) triangular solves with that factorisation (and its conjugate transpose);
(P3) dense arithmetic on objects of size r or 2 (the controller order and the coupling rank).

Nothing in the pipeline ever factorises, stores, or differentiates a matrix of closed-loop size. This is what "exploiting sparse + low-rank" means concretely, and it is the property that ports to the cavity, where (P1) becomes a sparse multifrontal factorisation of the linearised Navier–Stokes operator and (P2)–(P3) are unchanged.

1 The one structural fact

Plant (n complex states, M symmetric positive definite mass matrix, A sparse):

$$M\dot{q} = Aq + bu + d, \qquad y = c^{\mathsf{H}} q,$$

with actuator $b \in \mathbb{C}^n$, sensor $c \in \mathbb{C}^n$ (Gaussians of the positions, or free fields through a fixed smoothing map), and d the full-field disturbance. Controller (order $r = 8$; A_c tridiagonal with rows (d_c, e_c, f_c); input matrix B_c *fixed*; output row n^{T} and direct gain K free where the case allows):

$$\dot{x}_c = A_c x_c + B_c y, \qquad u = n^{\mathsf{T}} x_c + K y.$$

Substituting u and y, the closed-loop pencil on the stacked state $(q, x_c) \in \mathbb{C}^{n+r}$ is

$$M_{\mathrm{cl}} = \begin{bmatrix} M & 0 \\ 0 & I_r \end{bmatrix}, \qquad A_{\mathrm{cl}} = \underbrace{\begin{bmatrix} A & 0 \\ 0 & A_c \end{bmatrix}}_{\text{sparse} \oplus \text{small}} + \underbrace{\begin{bmatrix} b & 0 \\ 0 & B_c \end{bmatrix}}_{U \in \mathbb{C}^{(n+r)\times 2}} \underbrace{\begin{bmatrix} Kc^{\mathsf{H}} & n^{\mathsf{T}} \\ c^{\mathsf{H}} & 0 \end{bmatrix}}_{V \in \mathbb{C}^{2\times(n+r)}}.$$

$$\boxed{A_{\mathrm{cl}} \;=\; \mathrm{blkdiag}(A, A_c) \;+\; UV, \qquad \mathrm{rank}(UV) \le 2.} \tag{1}$$

Every formula below is a consequence of (1).

More actuators and sensors. The rank-2 structure is the 1×1 case of a general fact. With actuator matrix $B_a \in \mathbb{C}^{n\times n_a}$, sensor matrix $C_s \in \mathbb{C}^{n\times n_s}$ (so $y = C_s^{\mathsf{H}} q \in \mathbb{C}^{n_s}$, $u \in \mathbb{C}^{n_a}$) and a MIMO controller ($B_c \in \mathbb{R}^{r\times n_s}$, $N \in \mathbb{C}^{n_a\times r}$, $K \in \mathbb{C}^{n_a\times n_s}$), the coupling term of (1) becomes $U \in \mathbb{C}^{(n+r)\times(n_a+n_s)}$, $V \in \mathbb{C}^{(n_a+n_s)\times(n+r)}$: $\operatorname{rank}(UV) \le n_a + n_s$ — still tiny. Everything upgrades by replacing scalars with small matrices: the controller collapses to the $n_a \times n_s$ transfer *matrix* $\mathsf{K}(s) = K + N(sI - A_c)^{-1}B_c$; Sherman–Morrison becomes Woodbury with an $(n_a{+}n_s)$-sized capacity matrix; the plant transfer is $\mathsf{G}(s) = C_s^{\mathsf{H}} R(s) B_a \in \mathbb{C}^{n_s\times n_a}$ (n_a solves on one factorisation); and the winding scalar becomes the *return-difference determinant*, $f(s) = \det\big(I - \mathsf{K}(s)\mathsf{G}(s)\big)$ ($n_s \times n_s$, dense-tiny), with the factorisation (7) and the count formula (9) holding verbatim. Costs: the LU is unchanged; triangular solves multiply by n_a (forward) and n_s (adjoint); gradients still come from one forward/adjoint *pair of blocks* per frequency. The capability is scheduled (§12, item 7).

2 Every closed-loop solve is a plant solve

Let $s \in \mathbb{C}$ (on the frequency axis, $s = i\omega$) and consider the closed-loop system $(sM_{\mathrm{cl}} - A_{\mathrm{cl}})\binom{q}{x_c} = \binom{r_q}{r_c}$. Eliminate the controller states from the second block row: $x_c = (sI - A_c)^{-1}\big(r_c + B_c\, c^{\mathsf{H}} q\big)$ — an $r \times r$ dense solve, primitive (P3). Substituting into the first row collapses the whole controller to a single scalar, its transfer function at s:

$$\boxed{k(s) \;=\; K + n^{\mathsf{T}}(sI - A_c)^{-1}B_c,} \tag{2}$$

and the plant-level system becomes $\big(R(s)^{-1} - k(s)\, b\, c^{\mathsf{H}}\big)\, q = \tilde{r}$, with $R(s) = (sM - A)^{-1}$ the plant resolvent. A rank-one perturbation of a factorised matrix is solved by the Sherman–Morrison identity:

$$\boxed{\big(R^{-1} - k\, bc^{\mathsf{H}}\big)^{-1} = R + \frac{k\, Rb\, c^{\mathsf{H}} R}{1 - k\, G(s)}, \qquad G(s) = c^{\mathsf{H}} R(s)\, b.} \tag{3}$$

So one closed-loop solve costs: one sparse LU of $R(s)^{-1}$ (P1, done once per s and reused for everything), two or three triangular solves (P2: applying R to $\tilde{r}$ and to b; $G = c^{\mathsf{H}} Rb$ comes free from the second), and scalars (P3). Adjoint systems $(\,\cdot\,)^{\mathsf{H}} x = y$ use the *same* factorisation via transposed-conjugate triangular solves — no second LU.

Note the scalar in the denominator of (3),

$$f(s) \;=\; 1 - k(s)\, G(s),$$

the *return difference* of the SISO loop. Keep it in mind: it reappears as the star of the stability certificate in §7. (Numerically, $|f|$ small means the solve is ill-conditioned — which happens exactly when s approaches a closed-loop eigenvalue, as §7 makes precise.)

3 The H_∞ value: $\sigma_1(\omega)$ as an eigenvalue

The performance channel is full-field: disturbance d into the plant equation, output the plant state q measured in the energy norm $\|q\|_M^2 = q^{\mathsf{H}} M q$. With $\Pi = [\,I_n \;\; 0\,]$ the extractor of plant states, the (weighted) transfer operator at frequency ω is

$$T(\omega) \;=\; M^{1/2}\,\Pi\,(i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-1}\,\Pi^{\mathsf{T}} M^{1/2} \;\in\; \mathbb{C}^{n\times n}, \qquad \sigma_1(\omega) = \|T(\omega)\|_2.$$

We never form T; we need only its largest singular value, i.e. the largest eigenvalue of the Hermitian positive semidefinite operator $T^{\mathsf{H}}T$. Unwinding the square roots, σ_1^2 is the largest eigenvalue of

$$\boxed{W(\omega) \;=\; M R_{\mathrm{cl}}(\omega)^{\mathsf{H}} M R_{\mathrm{cl}}(\omega), \qquad R_{\mathrm{cl}}(\omega) = \Pi(i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-1}\Pi^{\mathsf{T}},} \tag{4}$$

understood as an operator on $\mathbb{C}^n$ (self-adjoint in the M^{-1} inner product; the code applies it verbatim: solve, multiply by M, adjoint-solve, multiply by M). One application of W = two solves of §2 on one factorisation. The largest eigenvalue is obtained by power/Lanczos iteration on W, *warm-started* with the converged eigenvector from the neighbouring frequency — across an exchange set the vectors vary slowly in ω, and the warm start cuts the iteration count to a handful. The converged eigenvector v (the worst disturbance shape) and its image give the singular pair (u, σ_1, v) of T that the gradients need next.

Cost of one oracle value: one LU + $O(10)$ triangular solves. On the 1-D GL mesh the LU is $O(n)$; at cavity scale it is the sparse factorisation of a shifted Navier–Stokes operator — expensive, which is why the outer method was shaped so that *many independent frequencies* are evaluated concurrently.

4 Gradients of σ_1: all parameters from one solve pair

Fix ω and suppose σ_1 is a *simple* singular value of T (the solver monitors the gap $\sigma_1 - \sigma_2$; more on ties in §8). With unit singular vectors u, v ($Tv = \sigma_1 u$), first-order perturbation theory gives the exact directional derivative

$$\delta\sigma_1 \;=\; \mathrm{Re}\big[\,u^{\mathsf H}\,(\delta T)\,v\,\big].$$

Since only A_{cl} depends on the design, $\delta\big((i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-1}\big) = (i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-1}\,\delta A_{\mathrm{cl}}\,(i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-1}$, and defining the two *solved vectors* (one forward, one adjoint solve — both on the factorisation already computed for the value):

$$\tilde v = (i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-1}\Pi^{\mathsf T} M^{1/2} v \in \mathbb{C}^{n+r}, \qquad \tilde u = (i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}})^{-\mathsf H}\Pi^{\mathsf T} M^{1/2} u \in \mathbb{C}^{n+r},$$

we obtain the master formula

$$\boxed{\;\delta\sigma_1 \;=\; \mathrm{Re}\big[\,\tilde u^{\mathsf H}\,(\delta A_{\mathrm{cl}})\,\tilde v\,\big].\;} \tag{5}$$

Everything now follows from the *structure* (1) of A_{cl}: each design parameter perturbs A_{cl} in a known sparse or low-rank pattern, so (5) evaluates to inner products of pieces of $\tilde u, \tilde v$. Write $\tilde v = (\tilde v_p, \tilde v_c)$ and $\tilde u = (\tilde u_p, \tilde u_c)$ for the plant and controller components. Then (for a complex parameter p we record the complex sensitivity ζ_p defined by $\delta\sigma_1 = \mathrm{Re}[\bar\zeta_p\,\delta p]$; the real gradient with respect to $(\mathrm{Re}\,p, \mathrm{Im}\,p)$ is $(\mathrm{Re}\,\zeta_p, \mathrm{Im}\,\zeta_p)$):

parameter	enters A_{cl} as	sensitivity from (5)
gain K	$\delta K\,[b;0][c^{\mathsf H}\;0]$	$\bar\zeta_K = (\tilde u_p^{\mathsf H} b)\,(c^{\mathsf H}\tilde v_p)$
output row n	$[b;0][0\;\;\delta n^{\mathsf T}]$	$\bar\zeta_{n_j} = (\tilde u_p^{\mathsf H} b)\,(\tilde v_c)_j$
A_c entries (d_c, e_c, f_c)	$\delta(A_c)_{jk}$ in the controller block	$\bar\zeta_{jk} = (\tilde u_c)_j^*\,(\tilde v_c)_k$ (tridiagonal jk only)
actuator b	$\delta b\,[Kc^{\mathsf H}\tilde v_p + n^{\mathsf T}\tilde v_c]$ direction	∇_b-vector $\propto\ \overline{\tilde u_p}\cdot\big(Kc^{\mathsf H}\tilde v_p + n^{\mathsf T}\tilde v_c\big)$
sensor c	$[\tilde u_p^{\mathsf H} bK + \tilde u_c^{\mathsf H} B_c]\,\delta c^{\mathsf H}$ direction	∇_c-vector $\propto\ \tilde v_p\cdot\overline{\big(K\,\tilde u_p^{\mathsf H} b + \tilde u_c^{\mathsf H} B_c\big)}$
position x_a	chain rule through $b = g(x_a)$	$\partial\sigma/\partial x_a = \mathrm{Re}\big[\,\zeta_b^{\mathsf H}\,g'(x_a)\,\big]$ (similarly x_s)

Free fields are optimised through a fixed linear smoothing (Sobolev) map $b = \Sigma\beta$; the chain rule multiplies ∇_b by the constant $\Sigma^{\mathsf H}$. The point to internalise is the **cost**: the entire gradient — all NZ real parameters, whether $\mathrm{NZ} = 2$ (a position case) or $\mathrm{NZ} = 1659$ (a two-field case) — costs *two* triangular solve sets beyond the value, independent of NZ. This is the adjoint-state economy, and it is why the field cases are affordable at all.

5 Frequency derivatives, peak location, and polishing

The same master formula with θ frozen and ω varied ($\delta(i\omega M_{\mathrm{cl}} - A_{\mathrm{cl}}) = i\,\delta\omega\, M_{\mathrm{cl}}$) gives

$$\frac{\partial\sigma_1}{\partial\omega} = \mathrm{Re}\big[\,\tilde u^{\mathsf H}(-iM_{\mathrm{cl}})\tilde v\,\big] = \mathrm{Im}\big[\,\tilde u^{\mathsf H} M_{\mathrm{cl}}\,\tilde v\,\big]\,:$$

one inner product per frequency, no extra solve. A peak is a zero of $\partial_\omega\sigma_1$; Newton's iteration on this scalar (second derivative by differencing the first, or one extra solve) polishes each located peak to 10^{-10} in a few oracle calls, warm-started along the iteration. Peak *location* starts from the exchange set and from a coarse scan; *missed* peaks are the certificate's job:

6 Certifying the global maximum

The optimiser must never trust a locally found maximum (v6, §5). The certified question is: *given a level γ, does $\sigma_1(\omega) < \gamma$ hold for all ω in the band?* The classical equivalence that turns this analytic question into an algebraic one: γ is a singular value of $T(\omega)$ if and only if the structured pencil

$$\boxed{\;\mathscr{P}_\gamma(\lambda) \;=\; \begin{bmatrix} -A & \gamma^{-1}\hat B\hat B^{\mathsf H} \\ -\gamma^{-1}\hat C^{\mathsf H}\hat C & -A^{\mathsf H}\end{bmatrix} - \lambda\begin{bmatrix} -M & 0 \\ 0 & M^{\mathsf H}\end{bmatrix}\;} \tag{6}$$

(written here for the closed-loop realisation, $\hat{B},\hat{C}$ the weighted input/output maps of §3) is singular at $\lambda = i\omega$. Consequently: if the pencil has *no eigenvalue* on a segment of the imaginary axis, then $\sigma_1 \neq \gamma$ throughout that segment, and a single sample below γ certifies the whole segment below γ. The oracle therefore covers the band by **eigenvalue-free disks**: at a test point $i\omega_0$, a structure-preserving shift-invert Arnoldi iteration finds the pencil eigenvalue nearest to $i\omega_0$; its distance is a certified radius within which the level is not crossed. Each shift-invert solve is a bordered sparse system that reduces, once again, to (P1)–(P3) — plant factorisations plus rank-$O(1)$ algebra. Adaptive placement of test points sweeps the band; wherever a disk cannot be certified, the iteration has *found* an axis-near eigenvalue, i.e. located a frequency where σ_1 reaches γ — exactly the "missed peak" the exchange needs. Bisection on γ (warm-started from the incumbent level) certifies the maximum itself to relative tolerance 10^{-6} at a cost of $O(10^2)$ shift-invert solves per design.

7 Stability: value, certificate, and the return difference

7.1 The value

The margin constraint is $\alpha(\theta) \le -\varepsilon$ with $\alpha = \max_j \operatorname{Re}\lambda_j(A_{\mathrm{cl}}, M_{\mathrm{cl}})$. During optimisation the pipeline does not recompute the whole spectrum; it *tracks* the few dangerous eigenvalues by shift-invert (inverse) iteration, shifting at each accepted step to the eigenvalue's previous location — each iteration a bordered solve on the plant factorisation. At benchmark scale a dense generalized eigensolve provides ground truth; at cavity scale it will not exist, which is why a certificate is required:

7.2 The certificate: count eigenvalues without computing them

Take determinants in the block form of $sM_{\mathrm{cl}} - A_{\mathrm{cl}}$ and eliminate the controller block (Schur complement), then apply the matrix determinant lemma to the remaining rank-one term — the same two moves as §2, now on determinants:

$$\begin{aligned}\det(sM_{\mathrm{cl}} - A_{\mathrm{cl}}) &= \det(sI - A_c)\,\det\big(R(s)^{-1} - k(s)\,bc^{\mathsf{H}}\big)\\ &= \det(sI - A_c)\;\det R(s)^{-1}\;\big(1 - k(s)\,G(s)\big).\end{aligned} \tag{7}$$

The passage from $R\,bc^{\mathsf{H}}$ to the scalar G is the *matrix determinant lemma*, in two lines. Factor out: $\det(R^{-1} - k\,bc^{\mathsf{H}}) = \det R^{-1}\cdot\det(I - k\,(Rb)\,c^{\mathsf{H}})$. Now $(Rb)\,c^{\mathsf{H}}$ is a rank-one matrix; its eigenvalues are $c^{\mathsf{H}}(Rb) = G$ (eigenvector Rb) and 0 with multiplicity $n-1$, so $\det(I - k\,(Rb)c^{\mathsf{H}}) = \prod_i(1 - k\lambda_i) = 1 - k\,G$. The row and column vectors trade places inside the determinant exactly because a rank-one matrix's only nonzero eigenvalue is the scalar "row times column".

$$\boxed{\det(sM_{\mathrm{cl}} - A_{\mathrm{cl}}) \;=\; \det R(s)^{-1}\,\det(sI - A_c)\,f(s), \qquad f(s) = 1 - k(s)G(s).} \tag{8}$$

The closed-loop spectrum is the union of the plant spectrum, the (open) controller spectrum, and the zeros of the scalar f — with the crucial feature that the poles of f are exactly the plant and controller eigenvalues, so the products cancel where they must.

Why the poles of f must sit exactly at the open-loop eigenvalues: count orders at one point. Let λ_p be a simple *plant* eigenvalue that is *not* a closed-loop eigenvalue. The left side $\det(sM_{\mathrm{cl}} - A_{\mathrm{cl}})$ is a polynomial, finite and nonzero at λ_p. On the right, $\det R(s)^{-1} = \det(sM - A)$ vanishes to first order at λ_p — so f must blow up to first order there to keep the product finite, and indeed it does: $G(s) = c^{\mathsf{H}}R(s)b$ has a simple pole at every plant eigenvalue (the resolvent blows up), hence so does $f = 1 - kG$. The same argument at controller eigenvalues, where $k(s)$ carries the pole. Nothing is being arranged; the factorisation (7) is an identity between a polynomial and a product of two vanishing factors and one blowing factor, and the orders must balance — which is precisely why the pole count $P_{\mathrm{plant}} + P_{\mathrm{ctrl}}$ appears in the count formula (9): the argument principle counts f's zeros *minus* its poles, and the poles are exactly the open-loop eigenvalues the two determinant factors account for. Now apply the argument principle to f on the boundary of the half-plane $\{\operatorname{Re}s > \ell\}$ (the vertical line $\operatorname{Re}s = \ell$ closed by a large arc): zeros minus poles of f inside equals the winding number of f along the contour, hence

$$\boxed{\#\{\lambda_{\mathrm{cl}} : \operatorname{Re}\lambda_{\mathrm{cl}} > \ell\} \;=\; W\big[f;\ \operatorname{Re}s = \ell\big] \;+\; P_{\mathrm{plant}}(\ell) \;+\; P_{\mathrm{ctrl}}(\ell),} \tag{9}$$

The winding number, defined. $W[f;\mathrm{Re}\,s=\ell]$ is the winding number of the curve $f(s)$ around the origin as s traverses the boundary of the region $\{\mathrm{Re}\,s>\ell\}$ with the region kept on the left: *down* the line, from $\ell+iY$ to $\ell-iY$, closed by the large arc through $\ell+Y$. Equivalently $W=\frac{1}{2\pi}[\arg f]_{\text{contour}}=\frac{1}{2\pi i}\oint f'/f\,ds$, and with this orientation the argument principle reads $Z-P=W$ over the region, which is how (9) is meant.

How it is computed: exact phase tracking, not quadrature. Numerical integration of f'/f would carry a quadrature error, and an error of 0.51 in a count is a wrong theorem. Instead the code tracks the phase: sample f along the segment, and between consecutive samples add the branch-consistent increment $\Delta\arg\in(-\pi,\pi]$; the sum telescopes to the exact total phase change *provided no true increment ever exceeded π unnoticed*. Three mechanisms enforce that proviso, each added after a measured miscount during validation: (i) *mandatory resolution near the poles of f* — near an open-loop eigenvalue close to the line the phase rotates fastest, so the sampler is forced to a density scaled by the distance to the nearest pole; (ii) *midpoint validation* — every accepted increment is re-checked at the interval's midpoint: the two half-increments must sum to the whole, a cheap proof that no $\pm2\pi$ was skipped; (iii) a *mandatory knot grid*, so that local adaptivity cannot glide over a feature between coarse seeds. With all three, the certificate reproduced the dense count on $104/104$ validation designs.

Parallel phase tracking. The tracked quantity is a sum of increments, and sums split: the segment is cut at the mandatory knots, each inter-knot interval goes to a worker that runs the serial predictor–validator march and returns its phase subtotal, and the subtotals add exactly — no phase continuity across cuts is needed, because no absolute branch of $\arg f$ is ever chosen (the only absolute arguments are at $\pm Y$, where f is confined near 1 and the principal branch is unambiguous). Two passes: knot values first (one parallel round); then intervals whose endpoint Hermite estimate already validates the chord increment are closed at zero extra cost, and the rest go to a work queue with midpoint splitting. The wall time is the longest interval — a few tens of serial samples, set by the hottest pole window — not the total count. The knot grid is 121 points over the core band, seven around every hot pole, plus the tail knots to $\pm Y$; shared-knot values are computed once in pass 1, so no duplicated factorisations; the three hardening mechanisms act *locally*, so they parallelise unchanged; and roundoff cannot flip a count, since an increment would have to move by π. Honest status: `winding_cert.py` as frozen is serial; the decomposition is §12, item 4, to be measured at GL where the serial reference exists.

The tail bound, derived slowly. The segment must end somewhere; the question is where the phase provably stops moving. Start from the Neumann series: for any square matrix B and $|s|>\|B\|$,

$$(sI-B)^{-1}=\frac{1}{s}\Big(I-\frac{B}{s}\Big)^{-1}=\frac{1}{s}\sum_{k\ge0}\Big(\frac{B}{s}\Big)^{k},\qquad\text{so}\qquad\|(sI-B)^{-1}\|\le\frac{1}{|s|}\cdot\frac{1}{1-\|B\|/|s|}=\frac{1}{|s|-\|B\|}.$$

Apply it to the plant resolvent by factoring out the mass matrix, $R(s)=(sM-A)^{-1}=(sI-M^{-1}A)^{-1}M^{-1}$: with

$$\nu:=\|M^{-1}A\|\quad\text{(any upper bound will do; a few power iterations on } M^{-1}A\text{, once per mesh)},$$

and $\|M^{-1}\|=1/\lambda_{\min}(M)$ (one Lanczos run on the symmetric positive definite M, once per mesh; for a lumped mass matrix, the smallest diagonal entry),

$$\|R(s)\|\;\le\;\frac{1}{\lambda_{\min}(M)\,(|s|-\nu)}\qquad\Longrightarrow\qquad|G(s)|=|c^{\mathsf H}R(s)b|\;\le\;\frac{\|c\|\,\|b\|}{\lambda_{\min}(M)\,(|s|-\nu)}.$$

Computing the once-per-mesh constants. Start from $\lambda_{\min}(M)=1/\lambda_{\max}(M^{-1})$: "Lanczos on M" in the form that actually converges quickly is shift-invert Lanczos with shift 0, i.e. the Lanczos (power) iteration on the operator $x\mapsto M^{-1}x$. Concretely: one sparse Cholesky of M — M is SPD, tridiagonal-banded on the 1-D mesh, a nested-dissection factor in 2-D — then each Lanczos step is two triangular solves, and the largest eigenvalue of M^{-1} is extremal, so it converges in a few tens of steps. Sparsity is exploited throughout: M is touched only through its factor. This is the one place the pipeline factorises the mass matrix, and it is admissible because it happens once per mesh, outside every loop, and the factor is discarded afterwards; the rule "never factorise M" (v2 report, §1.3) was about the optimisation loop. Now the honest accounting, in three thrift levels. *(a) What the shipped code does* (`winding_cert.py`, `plant_tail_constants`): `eigsh(M, k=1, which='SM')`, i.e. Lanczos on M itself in regular mode, asking ARPACK for the smallest-magnitude Ritz value — no factorisation at all, and fine at $n=399$, but it is the slow way (the small end of a mass-matrix spectrum is the clustered end, and regular-mode Lanczos converges to a clustered extremal eigenvalue in $O(\sqrt{\mathrm{cond}\,M})$ steps); at cavity scale use the

shift-invert form above, or better, (b). For ν the code runs the power iteration on $M^{-1}A$ with an LU of M — the same once-per-mesh exception. *(b) No eigensolve at all.* Only a *lower* bound on $\lambda_{\min}(M)$ is needed (a smaller value only pushes Y outward), and finite elements hand one over for free: with $M = \sum_e P_e^\top M_e P_e$ the assembly of element mass matrices, $x^{\mathsf{H}} M x = \sum_e x_e^{\mathsf{H}} M_e x_e \geq \sum_e \lambda_{\min}(M_e)\|x_e\|^2 \geq \big(\min_e \lambda_{\min}(M_e)\big)\|x\|^2$, since every degree of freedom belongs to at least one element; hence $\lambda_{\min}(M) \geq \min_e \lambda_{\min}(M_e)$ — a loop over elements with 3×3 (P2, 1-D) or 6×6 (P2, 2-D triangles) eigenproblems, no global solve, and certified. Likewise an *upper* bound on ν suffices: $\nu \leq \|A\|_2/\lambda_{\min}(M) \leq \sqrt{\|A\|_1\|A\|_\infty}/\lambda_{\min}(M)$, two column/row sums — pessimistic (it may push Y out by an order of magnitude), but Y only lengthens the quiet part of the tracked segment, which is cheap. *(c) Lumped mass*: the smallest diagonal entry, exactly.

Singular mass matrix: a real gap in the derivation above. The bound presumes M invertible, and the cavity's is not. The discretised incompressible Navier–Stokes operator is a differential-algebraic pencil: $M = \mathrm{diag}(M_v, 0)$ (mass on velocity, nothing on pressure), index 2. Three consequences. (i) $\lambda_{\min}(M) = 0$, M^{-1} does not exist, and the bound as written is void — it must be re-derived, not patched. (ii) The pencil has infinite eigenvalues (as many as the algebraic constraints), which are not eigenvalues of anything physical; $\det(sM - A)$ is a polynomial of degree below n, the argument principle for f still holds — f is a rational function whose finite poles are the finite plant and controller eigenvalues — but the tail closure is exactly the part that needs the re-derivation. (iii) The repair: the physics lives on the discretely divergence-free subspace. Writing the DAE as $M_v\dot{v} = A_{vv}v + B^\top p + b\,u$, $Bv = 0$, and eliminating p with the discrete Leray projector Π (the M_v-orthogonal projector onto $\ker B$; applying Π is one saddle-point, Stokes-type, solve), the velocity dynamics are an ODE, $\dot{v} = \Pi M_v^{-1} A_{vv} v + \Pi M_v^{-1} b\,u$, and for a channel that forces the momentum equation and measures velocity — which is our case: b is a body force (a boundary actuation is recast as one by lifting), c a velocity functional — $G(s)$ is strictly proper and the Neumann argument goes through verbatim on the subspace, with $\nu := \|\Pi M_v^{-1} A_{vv} \Pi\|$ (power iteration; each application one M_v solve plus one projection; once per mesh) and $\lambda_{\min}(M_v)$ of the velocity mass matrix, which is SPD, so (b) applies to it unchanged. Two design constraints fall out of this and are worth recording now. If the sensor measured *pressure* (a common cavity choice), G would be proper but not strictly proper — pressure answers a force instantaneously — so $G(\infty) \neq 0$ and $f(s) \to 1 - KG(\infty)$ rather than 1; the winding argument survives provided the tail confines f to a disk not containing the origin, i.e. $|KG(\infty)| < 1$ or a strictly proper controller ($K = 0$). And if an input entered the *constraint* equation, G would carry a polynomial part and no tail closure would exist — so inputs must stay in the momentum equation, which is a modelling rule, not a numerical one. The projected-subspace version of the bound is written up in §12 as the first correction to make, and it should be validated on a Stokes/lid-driven benchmark before the cavity.

7.3 Extension to Navier–Stokes: singular mass, real arithmetic

The extension to the cavity changes two things: the mass matrix becomes singular (descriptor form), and the arithmetic becomes real. Both are covered here; the shift-invert-at-zero decision for $\lambda_{\min}(M_v)$ is recorded in §12, item 1.

Descriptor structure (recap of the repair). The cavity pencil is $M = \mathrm{diag}(M_v, 0)$, index-2: the tail bound is re-derived on the discretely divergence-free subspace via the Leray projector Π, with $\nu = \|\Pi M_v^{-1} A_{vv} \Pi\|$ and $\lambda_{\min}(M_v)$; the modelling rules are *inputs in the momentum equation only* (else G grows a polynomial part and no tail closure exists) and *velocity outputs preferred* (pressure sensors leave G proper but not strictly proper, shifting the tail disk to $1 - KG(\infty)$, admissible only if $|KG(\infty)| < 1$ or $K = 0$). All once-per-mesh constants are computed on M_v, which is SPD, so every recipe of this section applies unchanged.

Real arithmetic (what changes, all of it in our favour). The GL plant is complex; the Navier–Stokes plant is *real*, and realness is structure: (i) *Conjugate symmetry of the spectrum.* Eigenvalues come in pairs $\lambda, \bar{\lambda}$; the rightmost eigenvalue is either real or a conjugate pair with *equal* real parts — so the abscissa "tie" created by a complex pair is not a kink at all (one constraint $\mathrm{Re}\,\lambda \leq -\varepsilon$ covers both members, since $\mathrm{Re}\,\lambda = \mathrm{Re}\,\bar{\lambda}$ identically), and per-eigenvalue tracking lists only the $\mathrm{Im} \geq 0$ representatives. (ii) *Frequency-response symmetry.* $\sigma_1(-\omega) = \sigma_1(\omega)$, so the band, the working set, the certification sweep and the peak polish all live on $\omega \geq 0$: half the frequencies, half the certification cost, for free. (iii) *Real factorisations.* Away from the frequency axis's complex shifts, the once-per-mesh objects (M_v, $A^{\mathsf{H}}A$, $\sigma M - S$) are real symmetric: real Cholesky/Lanczos, roughly a factor 2–4 cheaper in memory and flops than their complex counterparts. The frequency-axis LUs of $R(i\omega)^{-1}$ remain complex (the shift is imaginary), as before. (iv) *Winding symmetry.* $f(\bar{s}) = \overline{f(s)}$, so the

phase increments on the lower half-line mirror the upper ones: track $[\ell,\ \ell+iY]$ only and double it, adding the (real-axis) start point once — the certificate cost halves, again for free. None of these changes any statement in this note; they change constants, all downward. (Added to the to-do list's item 1 validation plan: check both symmetries explicitly at the Stokes benchmark, since they are also free *consistency tests* of the implementation.) The controller scalar obeys the same logic one size down: from (2), $|k(s)| \le |K| + \|n\|\,\|B_c\|/(|s| - \|A_c\|)$ for $|s| > \|A_c\|$. Both bounds decay like $1/|s|$, so their product can be driven below any threshold.

Choosing the endpoint Y, and why the answer is a certified integer. Choose the threshold $\frac{1}{2}$ and let Y solve the scalar inequality

$$\Big(\sup_{|s|\ge Y} |k(s)|\Big) \cdot \frac{\|c\|\,\|b\|}{\lambda_{\min}(M)\,(Y-\nu)} \;\le\; \frac{1}{2}$$

(monotone in Y; one bisection, once per design). Beyond the endpoints, and on the whole closing arc, $|f(s)-1| = |k(s)G(s)| < \frac{1}{2}$: the curve f is confined to the disk of radius $\frac{1}{2}$ around 1. Confinement does *not* mean f is constant there — it keeps moving — but a curve inside that disk stays in the half-plane $\{\mathrm{Re}\, f > \frac{1}{2}\}$, its argument is confined to $(-\pi/6, \pi/6)$, and a curve whose argument lives in a sector of width $< 2\pi$ *cannot wind around the origin*. So the arc's net phase contribution is not zero by magic; it is a number provably in $(-\pi/3, \pi/3)$. The total over the *closed* contour is exactly $2\pi\times$ an integer — winding numbers of closed curves are integers by topology, not by numerics — so the implementation takes the tracked segment phase $\Delta\varphi$, and returns the nearest integer to $\Delta\varphi/2\pi$: the rounding is *certified*, because the unaccounted arc contribution is provably less than a sixth of a turn, far below the half turn at which nearest-integer rounding could err. Yes: truly an integer, and the round is part of the proof, not a numerical apology.

On the cost of the factorisation itself: factoring a sparse matrix does create fill, but how much is a matter of ordering and dimension, not a catastrophe. In 1-D, M is banded and the factor is banded with *zero* fill — $O(n)$. In 2-D, with a nested-dissection ordering, the Cholesky factor of an n-node mesh carries $O(n\log n)$ nonzeros and $O(n^{3/2})$ flops — the same class as the LUs of $R(s)^{-1}$ the pipeline performs by the hundreds, so "ruins sparsity" is too strong: it *taxes* it, at the standard rate, and once per mesh. That said, for M specifically there is a cheaper road: the mass matrix is spectrally well conditioned (condition number $O(1)$–$O(10^2)$, mesh-quality dependent, not n-dependent), so conjugate gradients solve $Mx = b$ in a few tens of *matvecs* with no factor at all — and CG inside Lanczos-on-M^{-1} is a perfectly good once-per-mesh scheme. So the efficiency ladder is: lumped diagonal (free) $\to$ element bound (free, certified) $\to$ CG-based shift-invert (matvecs only) $\to$ Cholesky-based shift-invert (fastest per iteration, pays fill once). The element-wise bound, for its part, is unconditionally a lower bound. The two inequalities in its derivation are exact: $x_e^{\mathsf{H}} M_e x_e \ge \lambda_{\min}(M_e)\|x_e\|^2$ is the definition of a smallest eigenvalue, and $\sum_e \|x_e\|^2 \ge \|x\|^2$ holds because every global degree of freedom appears in at least one element (shared dofs are counted *more* than once, which only helps). No mesh assumption, no asymptotics — a certified floor.

Two decisions close this topic. $\lambda_{\min}(M)$: shift-invert (Cholesky- or CG-based) for the tight value, with the element bound retained as the certified cross-check — a tight constant shortens the tracked segment on every future design, so the one-time cost repays itself immediately. $\|A\|_2$: yes — Lanczos on the Hermitian $A^{\mathsf{H}}A$ (matvecs only, extremal target, a few dozen iterations) gives the true $\|A\|_2 = \sqrt{\mu_{\max}}$, tighter than $\sqrt{\|A\|_1\|A\|_\infty}$ by typically a small factor but occasionally much more on convection-dominated operators; then $\nu \le \|A\|_2/\lambda_{\min}(M)$, or, tightest of all, the direct power iteration on $M^{-1}A$ already described. The recipe recorded in §12 (item 2, updated): tight values by shift-invert/Lanczos, certified bounds (element-wise; Hölder) computed alongside as floors/ceilings — the pipeline uses the tight value and *checks* it against the bound, so a bug in either is caught by the other.

P_{ctrl} counts controller poles right of the line: $P_{\mathrm{ctrl}}(\ell) = \#\{\lambda \in \mathrm{eig}(A_c) : \mathrm{Re}\,\lambda > \ell\}$, an exact $r\times r$ computation. One nuance worth noticing: the production line is $\ell = -\varepsilon/2 < 0$, so a *stable but slow* controller pole with $\mathrm{Re}\,\lambda \in (-\varepsilon/2, 0)$ is counted in P_{ctrl} — and correspondingly contributes a pole of f inside the region — even though nothing is unstable. The count formula balances this automatically; the point is only that "poles right of the line" is not a synonym for "unstable poles". where P_{plant} (plant eigenvalues right of the line) is *design-independent* — computed once per mesh, and at cavity scale supplied from

Obtaining P_{plant} certified at cavity scale. A one-time cost per mesh and Reynolds number, amortised over the entire campaign; the recipe has three steps. (1) *Confine*: bound the region where right-of-line eigenvalues can live. The numerical range gives it cheaply: $\mathrm{Re}\,\lambda \le \max_x \mathrm{Re}(x^{\mathsf{H}}Ax)/(x^{\mathsf{H}}Mx)$, a Hermitian eigenvalue problem for the symmetric part (one Lanczos run), and similarly for $|\mathrm{Im}\,\lambda|$ — together a certified bounded rectangle to the right of the line. (2) *Count inside the rectangle*: a contour-integral eigenvalue count (Beyn/FEAST style:

the number of eigenvalues inside a contour equals the rank of a contour integral of the resolvent, estimated with a block of random probes and quadrature on the rectangle's boundary — every quadrature node is one plant factorisation, reused for all probes), cross-validated at two quadrature orders and two probe blocks. (3) *Polish*: each counted eigenvalue individually by shift-invert to machine precision, so the count is backed by exhibited eigenvalues. For a cavity at moderately supercritical conditions the physical count is small and known from the global-instability literature, so step (2) is a confirmation priced at $O(10^2)$ factorisations — once, not per design.

Step (1) deserves care, because it is easy to misread as the non-Hermitian pencil $Ax = \lambda Mx$ — it is not. It is a Hermitian problem. Write $S = \frac{1}{2}(A + A^{\mathsf{H}})$ for the Hermitian part; if $Ax = \lambda Mx$ then $\lambda = x^{\mathsf{H}}Ax/x^{\mathsf{H}}Mx$ and $\operatorname{Re}\lambda = x^{\mathsf{H}}Sx/x^{\mathsf{H}}Mx \le \mu_{\max}$, where $\mu_{\max}$ is the largest eigenvalue of the Hermitian pencil $Sx = \mu Mx$ (the numerical range, or field of values, of A in the M inner product). Hermitian pencils are where Lanczos is at home, in three respects. *Finding the top*: extremal eigenvalues converge first; run regular-mode Lanczos on $M^{-1}S$ (the once-per-mesh Cholesky of M again), or — faster when the top of the spectrum is clustered, which for a convection–diffusion operator it is, the top being the least-damped end — shift-invert at a shift σ to the *right* of the spectrum, where $\sigma M - S$ is positive definite, hence Cholesky-factorisable, and Lanczos on $(\sigma M - S)^{-1}M$ returns the eigenvalues nearest σ, i.e. the top ones, in a handful of steps. *Certifying the top*: Rayleigh–Ritz gives a Ritz value $\theta \le \mu_{\max}$ — a lower bound, the wrong side for confinement — but a certified *upper* bound costs one factorisation: pick $\sigma = \theta + \delta$ and attempt the sparse LDL^{H} of $\sigma M - S$; if every pivot is positive, Sylvester's inertia theorem says $\sigma M - S \succ 0$, hence $\mu_{\max} < \sigma$. *A Cholesky that succeeds is a proof.* The same two moves on the skew part $\frac{1}{2i}(A - A^{\mathsf{H}})$ bound $|\operatorname{Im}\lambda|$, and the rectangle is certified. Singular M: on the velocity subspace with Π, as in the previous answer. So step (1) is a rectangle with a certificate, and it answers the one question no mesh of shifts can — *how far right and how far up must we look?* The common practice of shift-invert on a mesh of imaginary-axis shifts is step (3) of this recipe, and it does yield the count *if* it finds every eigenvalue; what it cannot supply is the word "every". Shift-invert Arnoldi at σ returns the eigenvalues nearest σ, not a radius within which it has seen everything; two neighbouring shifts leave a gap about which the method is silent, and an eigenvalue far from any shift (large real part, or large imaginary part beyond the last shift) is simply invisible. Step (2) is what turns "we found K" into "there are K". On the fluid-mechanics habit: it is a reasonable one, because the physical modes are few, isolated and known in advance — and that is exactly the regime in which step (2) is a cheap confirmation rather than a search. There is also a safety net specific to our use of the number: P_{plant} is a constant added to every design's count, so an error in it is a constant offset in every certificate — detected by the first design whose stability is known by other means (the continuation gate, or at benchmark scale the dense eigensolve): a design known to be stable would read a nonzero count. Honest status: at GL scale the shipped `plant_pole_data` is dense (and refuses $n > 3000$, asking for P_{plant} and the near-line modes explicitly); steps (1)–(3) are the cavity recipe, written but not yet run.

The contour-counting idea has its own literature, added to the reference list: T. Sakurai & H. Sugiura, "A projection method for generalized eigenvalue problems using numerical integration", *J. Comput. Appl. Math.* 159 (2003) 119–128; E. Polizzi, "Density-matrix-based algorithm for solving eigenvalue problems", *Phys. Rev. B* 79 (2009) 115112 (the FEAST solver); W.-J. Beyn, "An integral method for solving nonlinear eigenvalue problems", *Linear Algebra Appl.* 436 (2012) 3839–3863. Yes, it is the argument principle, in matrix form. Scalar version: $\frac{1}{2\pi i}\oint \varphi'/\varphi\, dz = Z - P$ for a meromorphic φ. Matrix version: the resolvent $R(z) = (zM - A)^{-1}$ is meromorphic with simple poles at the eigenvalues (assumed simple) and residues $v_j w_j^{\mathsf{H}}/(w_j^{\mathsf{H}} M v_j)$, so by the residue theorem

$$\frac{1}{2\pi i}\oint_\Gamma (zM - A)^{-1} M\, dz \;=\; \sum_{\lambda_j \in \operatorname{int}\Gamma} \frac{v_j w_j^{\mathsf{H}} M}{w_j^{\mathsf{H}} M v_j} \;=\; P_\Gamma,$$

the spectral projector onto the eigenspace of the eigenvalues inside Γ, whose rank — or trace — is the count. And the trace is $\frac{1}{2\pi i}\oint \operatorname{tr}[(zM - A)^{-1}M]\, dz = \frac{1}{2\pi i}\oint \frac{d}{dz}\log\det(zM - A)\, dz$: literally the argument principle for $\varphi = \det(zM - A)$. That answers "what about zeros": for the determinant the zeros *are* the eigenvalues (with multiplicity) and there are no poles at all — $\det(zM - A)$ is a polynomial — so Beyn/FEAST counts zeros only, nothing to subtract. Contrast with f in (9): it has poles as well, because we divided the plant and controller determinants out to reach a scalar; that division is what buys a one-dimensional contour at the cost of two pole counts. For P_{plant} no such low-rank reduction exists, hence the matrix integral, computed with probes: for a random $Y \in \mathbb{C}^{n\times m}$ with m at least the expected count, $A_0 = \frac{1}{2\pi i}\oint R(z)MY\, dz$ has rank equal to the count (generically), and $A_1 = \frac{1}{2\pi i}\oint zR(z)MY\, dz$ then yields the eigenvalues themselves from the $m \times m$ pencil (A_1, A_0) restricted to the range — Beyn's method, so steps (2) and (3) can share the factorisations. Quadrature: Gauss or trapezoid on each edge of the rectangle; the error decays exponentially in the number of nodes for eigenvalues at a distance from the contour and slowly for eigenvalues near it — which is why the count is cross-validated at two orders, and why the rectangle's left edge should sit at $\ell - \delta$ rather than on the line, letting the polish of step (3) decide

the side of ℓ for anything found near it. Rigour, honestly: the rank of A_0 is decided from a singular-value gap, a numerical judgement, not a theorem; the standard of rigour is "count K at two quadrature orders and two probe blocks, then exhibit K eigenvalues inside with residuals at machine precision" — a certificate by exhibition and consistency, of the same kind as the disk sweep of §6.

7.4 Contour counting from scratch: the argument principle in matrix form

The construction, in five steps, assuming nothing.

Step 1: one eigenvalue, one integral. Let λ_0 be a simple eigenvalue of $Ax = \lambda Mx$ with right/left eigenvectors v, w. Near λ_0 the resolvent behaves like a simple pole with a rank-one residue: $R(z) = (zM - A)^{-1} = \dfrac{v\,w^{\mathsf{H}}}{(z-\lambda_0)\,w^{\mathsf{H}}Mv} + (\text{analytic near } \lambda_0)$. (To check the residue: multiply by $(zM-A)$ and use $Av = \lambda_0 Mv$, $w^{\mathsf{H}}A = \lambda_0 w^{\mathsf{H}}M$.) So integrating around a small circle Γ_0 enclosing only λ_0, the analytic part contributes nothing and the residue theorem leaves $\frac{1}{2\pi i}\oint_{\Gamma_0} R(z)M\,dz = \dfrac{v\,w^{\mathsf{H}}M}{w^{\mathsf{H}}Mv}$ — a rank-one matrix that maps any vector onto the eigendirection v.

Step 2: many eigenvalues. For a contour Γ enclosing eigenvalues $\lambda_1, \ldots, \lambda_K$, residues add:

$$P_\Gamma := \frac{1}{2\pi i}\oint_\Gamma R(z)M\,dz = \sum_{j\le K} \frac{v_j w_j^{\mathsf{H}} M}{w_j^{\mathsf{H}} M v_j}.$$

This P_Γ is a *projector* ($P_\Gamma^2 = P_\Gamma$: multiply two terms and use the biorthogonality $w_i^{\mathsf{H}}Mv_j = 0$ for $i \neq j$): it projects onto the span of the enclosed eigenvectors.

Step 3: why trace = rank = count. A projector's eigenvalues are only 0 and 1 (any eigenvalue π of P obeys $\pi^2 = \pi$). Its *rank* is the dimension of the space it projects onto — the number of enclosed eigendirections, i.e. the count K. Its *trace* is the sum of its eigenvalues — K ones plus zeros $= K$ again. Trace and rank are two roads to the same integer; the trace is the analyst's road, the rank the algorithm's (step 5).

Step 4: the trace is the scalar argument principle in disguise. Jacobi's formula — the derivative of a determinant — says $\frac{d}{dz}\det X(z) = \det X(z)\cdot \operatorname{tr}\left[X(z)^{-1}X'(z)\right]$; proof in two lines: $\det(X+\varepsilon X') = \det X\,\det(I+\varepsilon X^{-1}X') = \det X\left(1+\varepsilon\operatorname{tr}[X^{-1}X'] + O(\varepsilon^2)\right)$, because $\det(I+\varepsilon B) = \prod_i(1+\varepsilon\beta_i) = 1+\varepsilon\sum_i\beta_i + O(\varepsilon^2)$. With $X = zM - A$, $X' = M$: $\operatorname{tr}[R(z)M] = \frac{d}{dz}\log\det(zM-A)$ — and it identifies $\operatorname{tr}P_\Gamma$ with $\frac{1}{2\pi i}\oint \varphi'/\varphi\,dz$ for $\varphi = \det(zM-A)$: the scalar argument principle, counting the polynomial's zeros (the eigenvalues, with multiplicity), no poles to subtract.

Step 5: computing without the trace — probes. Forming $\operatorname{tr}[R(z)M]$ exactly would need n solves per quadrature node. Instead hit the projector with a thin random block $Y_p \in \mathbb{C}^{n\times m}$, $m \gtrsim K$: $A_0 = P_\Gamma Y_p$ (per node: one factorisation, m solves). Since P_Γ has rank K, generically A_0 has rank exactly K, read off from its singular values (a clean gap after the K-th). That is the *count*. For the eigenvalues themselves, one more moment, $A_1 = \frac{1}{2\pi i}\oint zR(z)MY_p\,dz$, and the tiny $m\times m$ pencil built from (A_1, A_0) has the enclosed λ_j as its eigenvalues — Beyn's method.

Beyn or FEAST? Same integral, different engines. BEYN (2012) is the *direct* method just described: two moments, one SVD, one tiny pencil — non-iterative, ideal when K is small and one wants count-plus-eigenvalues in one shot. FEAST (Polizzi 2009 — the proper name of his solver and library, not an acronym) uses the same contour integral as a *filter* inside subspace iteration: apply P_Γ repeatedly to refine a subspace, Rayleigh–Ritz inside — iterative, more robust to quadrature error and to underestimating m, preferred when many interior eigenvalues are wanted. For P_{plant} — few eigenvalues, one shot, count first — **Beyn is the right tool**, and the to-do item now says so (§12, item 3).

Placing the contour. ℓ is the counting line's real part — the same ℓ as in (9); production value $\ell = -\varepsilon/2$. δ is the standoff of the rectangle's *left* edge from that line: the edge sits at $\ell - \delta$ so that near-line eigenvalues fall strictly *inside* the contour rather than on it, and the shift-invert polish decides on which side of ℓ each of them lies. *The contour*: right and top edges from the certified numerical-range rectangle (inertia certificate) plus a margin; bottom edge its mirror (or the real axis for a real plant, §7.3); left edge at $\ell - \delta$. *Y and ν, unified*: yes, the same family — ν confines the whole spectrum to the disk $|\lambda| \le \nu$ (a circumscribed-disk statement), while the numerical-range rectangle is the sharpened, per-direction version of the same confinement; the rectangle's $|\mathrm{Im}|$ bound can therefore *tighten* the winding endpoint Y too, and the cleanest implementation computes the rectangle once and derives both the contour and Y from it, keeping the Neumann-ν formula as the fallback when only the

crude constants exist. *Quadrature robustness*: the danger: an eigenvalue (or a region of large resolvent norm, the pseudospectrum of a non-normal A) close to an edge slows the quadrature's convergence. Three defences, layered: (i) *certified standoff*: sweep each contour edge with the eigenvalue-free disks of §6 — a 1-D sweep per edge, the same tool as the frequency band — which certifies a minimum spectral distance from the contour and, if a disk refuses to certify, moves the edge before any quadrature is attempted; (ii) *resolvent monitoring*: the quadrature already evaluates $R(z)MY_p$ at every node — its norm is read off for free, and a spike flags a near-pseudospectral passage even when eigenvalues are distant (non-normality), triggering edge displacement or local refinement; (iii) the *two-order cross-validation* of the count, which is the net under everything. Extra margin: yes, cheap and wise — the rectangle is enlarged by a fixed fraction before quadrature, since a larger contour costs nothing (the count inside is unchanged) while a tight one courts exactly that slow-convergence risk.

Why a mesh of shifts is not a certificate — and how to make it one. Shift-invert on a mesh of imaginary-axis shifts (perhaps with a second line at $\mathrm{Re}\, s = 1$) is what everyone does, and what the shipped code effectively does at GL scale (with a dense solve standing in for the mesh); it is not a certificate — for one reason that step (1) fixes and one that step (2) fixes. Without (1) the mesh has no *edges*: nothing establishes that the shifts need not go beyond $|\mathrm{Im}\, s| = 60$ or $\mathrm{Re}\, s = 1$, and a spurious or physical mode at $(2, 200)$ is invisible to both lines. The numerical-range rectangle is the only cheap object that answers "how far", and it is one Hermitian eigensolve plus one Cholesky. Without (2) the mesh has no *radii*: shift-invert at σ sees the nearest eigenvalues, and says nothing about the strip between two neighbouring shifts, so a second line at $\mathrm{Re}\, s = 1$ does not confirm anything — it adds a second row of blind spots, at half the distance. There is, however, a way to make exactly that scheme rigorous, with a tool we already own: the eigenvalue-free disks of §6. At shift σ, the distance $r(\sigma)$ to the nearest eigenvalue certifies the open disk of radius $r(\sigma)$ empty (with the same hedge as in §6: request a few eigenvalues, so that Arnoldi's "nearest" is trustworthy); cover the rectangle by such disks and the count is the number of eigenvalues exhibited. That is the Schwerdtner sweep run over an *area* instead of a line, and its cost is the catch: the number of disks grows like $(\text{area})/r^2$, and near the line the radii are tiny (the leading GL mode sits 0.007 from the production line), so the hot regions eat factorisations — whereas the contour count of (2) costs $O(10^2)$ factorisations regardless of where the eigenvalues sit. Verdict: keep (1), because nothing else bounds the search; take (2) as either the contour count or the disk covering, whichever one trusts more — both are the argument principle, one on the boundary and one by exhaustion of the interior; and use the mesh, and the second line, as what they are good at: shifts for the polish of (3), the second line being a sensible place to polish anything the count locates far to the right. And the fluid-mechanics practice of trusting the mesh is, seen this way, a covering argument supplied by physical prior knowledge — legitimate for a plant one knows, which is exactly why P_{plant}, computed once and cross-checked once, is the least worrying constant in the whole certificate.

How many eigenvalues per shift (`nev`). The campaign measured `nev` $= 1$ adequate *at GL scale under the validation harness*: every disk cross-checked against dense ground truth, $104/104$ — so "`nev` $= 1$ is enough here" is a measured fact, and paying $3\times$ per shift for hypothetical robustness was rightly refused. The opposite hedge — request a few — addresses the regime without ground truth (cavity), where a misconverged "nearest" silently inflates a disk. The shipped-and-recommended policy: `nev` $= 1$ by default, with *escalation on suspicion* — re-solve with `nev` $= 3$ only when the returned pair's residual is loose, when two adjacent disks disagree about their overlap, or when the disk is critical (its radius decides a certificate rather than merely advancing the sweep). Cheap where it can be, careful where it must be; at GL the escalation almost never fires, which is itself the measurement that justifies the default. the known physical instability structure — and P_{ctrl} is an exact $r \times r$ computation. Each sample of f along the line costs one plant LU ($G = c^{\mathsf{H}} R b$); f' costs one extra solve on the same factorisation ($G' = -c^{\mathsf{H}} R M R b$), enabling phase-tracking with derivative information (its use in practice: the question below). The arc at infinity is closed analytically by the tail bound derived above.

Using the derivative information. Two uses. *Predictor*: the local phase speed is $\mathrm{d}\arg f/\mathrm{d}s = \mathrm{Im}(f'/f)$, so the next sample is placed at $\Delta s \approx \varphi_{\text{budget}}/|f'/f|$ with a budget of a fraction of π — adaptive stepping by prediction instead of by trial-and-refine. *Validator*: the tracked increment is cross-checked against the Hermite estimate $\frac{\Delta s}{2}[\mathrm{Im}(f'/f)_{\text{left}} + \mathrm{Im}(f'/f)_{\text{right}}]$; disagreement triggers refinement before the midpoint test even runs. Cost: f' is one extra triangular solve on the *same* factorisation — at GL scale that is a $\sim 30\%$ overhead per sample, at cavity scale (where the LU dominates the solve by an order of magnitude) nearly free — against which the predictor cuts the sample count by a factor that was worth it at GL already and is decisive wherever each sample carries a fresh factorisation. So: worth it, increasingly so with scale. Sampling is adaptive with three hardening mechanisms (mandatory resolution near the poles of f, midpoint validation of accepted steps, and a

mandatory knot grid), each added after a measured failure; the certificate reproduces dense eigenvalue counts exactly on $104/104$ validation designs at

Cost at cavity scale, honestly. At cavity scale one factorisation is the dominant unit (minutes, gigabytes), so 500–900 of them serially is hours — and three facts make it acceptable and a fourth makes it smaller. (1) Cadence: the winding certificate runs per *saved* design — a handful per case — never per step. (2) Parallelism: samples are independent; 64 workers make it ~ 10 rounds, and the memory footprint is one factorisation *per worker at a time* (each worker processes its queue serially and discards), not one per sample — the footprint objection dissolves. (3) The derivative-informed predictor above cuts the sample count. (4) Structural reduction, if still needed: the samples concentrate near the poles of f; away from the hot regions the same Neumann logic that closes the arc certifies long quiet stretches of the *segment* from a few samples with interval bounds on $|kG|$ — a coarse-certify/fine-track split that brings the count toward $O(10^2)$. Projection, to be validated at GL first (to-do list of the optimisation note): tens of minutes per saved design on a modest allocation — a final-answer cost, not a loop cost.

$\approx$ 500–900 plant solves per design. The production line is $\ell = -\varepsilon/2$: a certified count of zero there is an *unconditional* proof of margin-stability, with no continuation premise.

8 Gradient of the abscissa — and do we need its subgradients?

8.1 The gradient

Let λ be the rightmost eigenvalue, assumed *simple*, with right and left generalized eigenvectors $A_{\mathrm{cl}} v = \lambda M_{\mathrm{cl}} v$ and $w^{\mathsf{H}} A_{\mathrm{cl}} = \lambda\, w^{\mathsf{H}} M_{\mathrm{cl}}$. Standard eigenvalue perturbation gives

$$\boxed{\delta\lambda \;=\; \frac{w^{\mathsf{H}}\,(\delta A_{\mathrm{cl}})\,v}{w^{\mathsf{H}} M_{\mathrm{cl}}\,v}, \qquad \delta\alpha = \operatorname{Re}\delta\lambda,} \tag{10}$$

which has *exactly* the shape of the master formula (5) with $(\tilde{u},\tilde{v})$ replaced by $(w/\overline{(w^{\mathsf{H}}M_{\mathrm{cl}}v)},\, v)$: the same group-by-group table of §4 applies verbatim, so the full gradient of α with respect to every design parameter costs *one eigenpair* — obtained by shift-invert iteration and its adjoint on one plant factorisation. (The denominator $w^{\mathsf{H}}M_{\mathrm{cl}}v$ also serves as a health monitor: it tends to zero when the eigenvalue approaches a defective (Jordan) configuration, where α ceases to be Lipschitz; non-generic, but worth watching.)

8.2 Do we need subgradients?

Three answers, in increasing order of care.

(i) Generically, no. For a complex (non-real) system like ours there is no automatic conjugate pairing, so the rightmost eigenvalue is generically unique and simple; α is then differentiable and (10) is its honest gradient. This was the measured situation even in the hardest case of the campaign: the design pinned on the margin (K+B+C) carried a *single* eigenvalue at the line — we tested the multiple-eigenvalue hypothesis explicitly and rejected it.

(ii) At ties, the subdifferential exists but we avoid needing it. $\alpha = \max_j \operatorname{Re}\lambda_j$ is a max, so when several eigenvalues tie at the rightmost real part the correct first-order object is the convex hull of their individual gradients (10) — structurally identical to the frequency-max story of the optimisation note. And the remedy is the same for the third time: *never take the max; name the members*. The diagnostic KKT tool already treats each near-line eigenvalue as its own function (gradients by matched perturbation of the tracked eigenvalues, multipliers by non-negative least squares); the planned explicit-constraint inner does likewise, imposing $\operatorname{Re}\lambda_j(\theta) \le -\varepsilon$ *per tracked eigenvalue*. Each such constraint is smooth with gradient (10); the tie lives in the active set, where the solver is competent, and the subdifferential of the max is never formed.

(iii) The genuinely nonsmooth points are defective eigenvalues, where two eigenvalues collide into a Jordan block and α behaves like a square root (non-Lipschitz; no useful subgradient). These are avoided rather than modelled: the margin formulation keeps the optimiser ε away from the axis where the physical coalescences live, and $|w^{\mathsf{H}}M_{\mathrm{cl}}v|$ flags any approach.

The parallel with the singular-value story is close but not exact, and the difference is worth owning. The exact analogue of the $\sigma_1 = \sigma_2$ crossing is case (ii) above — a *semisimple* tie, a max of two smooth branches,

a Lipschitz kink, handled by enrollment. Case (iii) — the defective collision — is strictly *worse* than anything singular values can do, and here is the pleasant reason why: the σ's are eigenvalues of the *Hermitian* operator $T^{\mathrm{H}}T$, and Hermitian eigenvalues are always semisimple — crossings, yes; Jordan blocks, never. Eigenvalues of the non-normal A_{cl} enjoy no such protection, which is exactly why the abscissa needs the extra care of case (iii) while the frequency response never does. Non-normality, the physical heart of this whole flow problem, is also the algebraic reason its stability constraint is the more delicate object.

In short: **value + gradient of individually tracked eigenvalues suffice**; the max structure is handled by enrollment, exactly as σ_2-enrollment handles singular-value ties, and no subdifferential machinery is required anywhere in the design.

9 How §5–§6 compare with Benner–Mitchell (2018)

The comparison is natural because their paper and our oracle share an architecture. Their starting point is the classical level-set method of Boyd–Balakrishnan and Bruinsma–Steinbuch ("BBBS", 1990): bisect on the level γ, and at each level pay one *full Hamiltonian eigendecomposition* to find all frequencies where $\sigma_1 = \gamma$ — robust, global, and $O(n^3)$ *per level*. Their observation: that work distribution is upside-down — cheap local information is being bought with the most expensive global tool. Their method rebalances: run standard *optimization* in ω to convergence on local peaks — this is where a reader may recall "splines": the local maximisation is accelerated by cubic/Hermite *interpolation* of $\sigma_1(\omega)$ using values and derivatives (a refinement lineage their paper traces back through the level-set literature), giving superlinear local convergence for a few evaluations — and invoke the expensive global eigendecomposition only to *verify* the converged candidate (typically once or twice per norm computation instead of once per bisection level). Faster, and, as their title says, more accurate: the optimizer polishes to full precision, which plain level-set bisection does not. Now place §5–§6 beside it. The division of labour is *identical* — optimise locally, certify globally, and arrange for the certificate to be consulted as rarely as possible: our Newton polish in ω (§5) is their interpolation-accelerated local stage (same derivative information, same superlinear character), and our certified disk sweep (§6) plays their verification

Our local stage differs from theirs in one respect: we use Newton on $\partial_\omega\sigma_1$. The two are close relatives: Hermite interpolation through $(\sigma_1, \partial_\omega\sigma_1)$ at two bracketing points is a secant-flavoured Newton that borrows information across the bracket, while our Newton uses one point and a second derivative. Asymptotically Newton is faster (quadratic versus superlinear); far from the peak the interpolation is more robust (it cannot propose a step outside its bracket). The choice is dictated by the starting point each method actually has: they start cold, from level-set midpoints, so robustness wins; our peaks are warm-started from the previous outer round, where the peak has barely moved, so Newton sits inside its basin from the first iterate and the quadratic rate is real. Both spend 2–4 oracle calls per peak; the difference is marginal, the contexts decide, and a safeguarded hybrid (Newton with a Hermite-bracket fallback when $\partial_\omega^2\sigma_1 \geq 0$ signals a plateau) is the best of both — noted for the next iteration. stage. The two differences are deliberate. *First, the global tool itself*: their verifier is the dense Hamiltonian eigendecomposition — exact, but $O(n^3)$ and dense-only; ours is the eigenvalue-free-disk certificate of Schwerdtner–Mengi–Voigt, which needs only sparse shift-invert solves, i.e. (P1)–(P3), and therefore survives the trip to cavity scale. So: no, they do not use the Schwerdtner-style certification — at their scales they do not need it; we adopt it *because* their verifier is the one component of the architecture that does not scale. *Second, the amortisation horizon*: they optimise and verify within *one* norm computation; our exchange loop amortises across an entire *design* trajectory — the working set Ω carries candidate peaks from design to design, warm-starting both the local stage (in ω *and* in θ) and the bisection of the certificate from the incumbent level. In one sentence: *our oracle follows the Benner–Mitchell division of labour — cheap local optimisation, rare global verification — with Newton in place of their interpolation, their dense verifier replaced by the Schwerdtner–Mengi–Voigt certificate, and the amortisation horizon stretched from one norm computation to the whole optimisation*.

("Per-norm" means per computation of one design's H_∞ norm; "amortise" means: spread the cost of the expensive component over many uses of its result. They amortise *within* one norm computation — one expensive global verification serves many cheap local steps. We keep that, and add a second horizon: *across designs* — the working set Ω, the singular vectors, the peak locations and the incumbent level all carry over from one design to the next, so the k-th norm evaluation starts where the $(k-1)$-th finished and costs a fraction of a cold one. On "architecture": fair objection — the word claimed too much, since our local stage is Newton, not their interpolation; the sentence above now says "division of labour" and names the substitution explicitly.

References. Boyd & Balakrishnan (1990); Bruinsma & Steinbuch (1990) — the BBBS level-set method. Guglielmi, Gürbüzbalaban & Overton, SIMAX 2013 — spectral-value-set iteration, the fast local (non-certified) large-scale line. Benner & Voigt, *Math. Control Signals Syst.* 2014 — structured pseudospectral H_∞ computation for large-scale descriptor systems. Aliyev, Benner, Mengi, Schwerdtner & Voigt, SIMAX 2017 — greedy subspace method for large-scale L_∞ norms. Benner & Mitchell, SISC 2018 — the optimization-rebalanced norm computation above. P. Schwerdtner, E. Mengi & M. Voigt, "Certifying global optimality for the L_∞-norm computation of large-scale descriptor systems", *IFAC-PapersOnLine* 53(2) (2020), 21st IFAC World Congress — the eigenvalue-free-disk certificate of §6; note that Mengi is a co-author of both this certificate and the subspace methods of §10, which share the same view of the problem. Apkarian & Noll, IEEE TAC 2006 (with the multiband and 2015 parametric papers, and Apkarian's 2018 infinite-dimensional structured H_∞) — the synthesis line, treated at length in the optimisation note. Curtis, Mitchell & Overton 2017 — the inner solver.

References for the contour counting. Sakurai & Sugiura, *J. Comput. Appl. Math.* 159 (2003) 119–128; Polizzi, *Phys. Rev. B* 79 (2009) 115112 (FEAST); Beyn, *Linear Algebra Appl.* 436 (2012) 3839–3863 — contour-integral eigenvalue counting and extraction, the matrix form of the argument principle used for P_{plant} in §7.2.

10 Projection: a reduced model for the inner loop

Everything before this section evaluates the oracle by one sparse factorisation per frequency, and the cost accounting of §11 is built on that unit. This section describes a different organisation of the same algebra — a low-dimensional model of the resolvent that serves every frequency, and every design, from a handful of factorisations — states what has been measured on GL, and what remains to be measured on the cavity. It rests on a published method, which is described first.

10.1 The method of Aliyev, Benner, Mengi, Schwerdtner and Voigt (2017)

Reference: N. Aliyev, P. Benner, E. Mengi, P. Schwerdtner, M. Voigt, "Large-scale computation of $\mathscr{L}_\infty$-norms by a greedy subspace method", *SIAM J. Matrix Anal. Appl.* 38(4) (2017) 1496–1516 (arXiv:1705.10086). The setting is more general than a descriptor system: an $\mathscr{L}_\infty$-function

$$H(s) = C(s)\,D(s)^{-1}B(s), \qquad D(s) = \sum_{j=1}^{\kappa_D} h_j(s)D_j, \quad B(s) = \sum_j f_j(s)B_j, \quad C(s) = \sum_j g_j(s)C_j,$$

with $D_j \in \mathbb{C}^{n\times n}$ large and sparse, B_j tall-and-skinny, C_j short-and-fat, and f_j, g_j, h_j meromorphic scalar functions — descriptor and delay systems are the paper's examples. *Our closed loop is natively of this form*: by §2, $D(s) = sM - A - k(s)\,bc^{\mathsf{H}}$, i.e. $\kappa_D = 3$ with $h = (s, 1, -k(s))$ and $D = (M, A, bc^{\mathsf{H}})$, the controller entering only through the scalar meromorphic coefficient $k(s)$ of (2). Note the $\mathscr{L}_\infty$ in the title: the supremum of σ_1 over the imaginary axis is defined whether or not the system is stable, and the method never assumes stability.

Reduction. Choose $V, W \in \mathbb{C}^{n\times\tilde{n}}$, $\tilde{n} \ll n$, and project the middle factor two-sidedly (Petrov–Galerkin): $\tilde{H}(s) = \tilde{C}(s)\tilde{D}(s)^{-1}\tilde{B}(s)$ with $\tilde{D}_j = W^{\mathsf{H}}D_jV$, $\tilde{B}_j = W^{\mathsf{H}}B_j$, $\tilde{C}_j = C_jV$. For us,

$$\tilde{D}(s) = s\,(W^{\mathsf{H}}MV) - (W^{\mathsf{H}}AV) - k(s)\,(W^{\mathsf{H}}b)(c^{\mathsf{H}}V)\,:$$

the two large projected matrices are computed once; the design enters only through the scalar $k(s)$ and the two small vectors $W^{\mathsf{H}}b$, $c^{\mathsf{H}}V$. This is the formal reason for the design-independence measured below.

Interpolation (their Theorem 2.1, a special case of Beattie–Gugercin 2009). At a point μ and for vectors b_0, c_0: if $D(\mu)^{-1}B(\mu)b_0 \in \mathrm{Col}(V)$ then $\tilde{H}(\mu)b_0 = H(\mu)b_0$; if $(c_0^{\mathsf{H}}C(\mu)D(\mu)^{-1})^{\mathsf{H}} \in \mathrm{Col}(W)$ then $c_0^{\mathsf{H}}\tilde{H}(\mu) = c_0^{\mathsf{H}}H(\mu)$; and if both, $c_0^{\mathsf{H}}\tilde{H}'(\mu)b_0 = c_0^{\mathsf{H}}H'(\mu)b_0$. Applied to the right/left singular vectors v, w of $\sigma_1(H(\mu))$, this gives *Hermite interpolation of the largest singular value*: σ_1 and $\partial_\omega\sigma_1$ of the reduced function equal those of the full function at μ. The statement is *per vector*, which matters for us below. Two consequences they draw: with only the input direction in V one has $\sigma_1(H(\mu)) \le \sigma_1(\tilde{H}(\mu))$ — the two-sided reduced function is an *interpolant*, not a bound; and V, W must have equal dimension for $\tilde{D}$ to be square.

Algorithm 1 (greedy). Start with one interpolation point $i\omega_1$ and the subspaces $V_1 = D(i\omega_1)^{-1}B(i\omega_1)$, $W_1 = (C(i\omega_1)D(i\omega_1)^{-1})^{\mathsf{H}}$. Then repeat: compute $\omega_r = \arg\max_\omega \sigma_1(\tilde{H}_{r-1}(i\omega))$ by a dense method on the reduced function; expand V, W with the snapshots at $i\omega_r$; orthonormalise. Each iteration costs one sparse factorisation of $D(i\omega_r)$ and $m + p$ solves. *Convergence (their Theorem 3.3)*: under a local convergence assumption the iterates

converge to a local maximiser of σ_1 *superlinearly in the subspace dimension*, and the proof uses only the Hermite interpolation at the last two iterates; keeping all previous subspaces (their Remark 2.3) costs little and gives much better global behaviour in practice, typically fewer than ten iterations to 10^{-6}. *Limitation (their §4.4, stated by the authors)*: convergence is local only, as for every large-scale $\mathscr{L}_\infty$ method to date; if the initial interpolation points miss the part of the axis near a pole, the global maximiser is missed — their own example converges to $\omega = 7.897$ when the maximiser is at 1.092. Their remedy is more initial points; there is no certificate. That limitation is exactly what §6 supplies, which is the point of §10.5.

10.2 Our variant: full-field channel, dominant directions, Woodbury columns

Our performance channel is not tall-and-skinny: $B = C^{\mathsf{H}} = M^{1/2}$ is $n \times n$, so $D^{-1}B$ is the whole space. Theorem 2.1 being per vector, the remedy is the reduced-basis variant the authors themselves indicate: at each anchor $i\omega_a$ include only the images of the *dominant* singular directions,

$$RM^{1/2}v_j \text{ in } V, \qquad R^{\mathsf{H}}M^{1/2}u_j \text{ in } W, \qquad j = 1, \ldots, n_{\mathrm{vec}},$$

where (u_j, σ_j, v_j) are the top singular triplets of $T(i\omega_a)$ — which the warm-started power/Lanczos iteration of §3 produces anyway — plus the two *Woodbury columns* Rb and $R^{\mathsf{H}}c$, so that the rank-one controller correction is representable inside the subspace exactly. With $n_{\mathrm{vec}} \geq 2$ the second singular value is Hermite-interpolated as well. Two projections are then in use: the *two-sided* one (interpolant, exact at anchors, no bound), for values, σ_2, gradients and the eigenvalue predictor; and the *one-sided input restriction* $\|TV\|$ with V orthonormal, which is a *certified lower bound* $\leq \sigma_1$ at every frequency, since restricting the input can only lower the gain. Gradients come from the master formula (5) with the solved vectors replaced by their subspace representatives $V\tilde{v}$, $W\tilde{u}$ — no full solve.

10.3 What was measured on GL

All tests dense, GL with $n = 799$ (or $n = 399$ where noted), performance channel $T(\omega) = M^{1/2}R_{\mathrm{cl}}(\omega)M^{1/2}$, $n_{\mathrm{vec}} = 3$, greedy anchors, against a dense ground truth.

Plant sweep ($\omega \in [-2, 12]$, 141 points): anchors $1/2/4/5$ give maximum relative error $7 \cdot 10^{-1}$ / $4.5 \cdot 10^{-1}$ / $7.8 \cdot 10^{-2}$ / $1.7 \cdot 10^{-4}$ over the band; the peak is captured to 10^{-6} from the second anchor on, and the greedy sequence chose the peak frequency second. Five factorisations in place of 141.

Closed loop at a fixed design (Woodbury form, $r = 8$ tridiagonal controller): 5 anchors, $\dim V = 40$, maximum relative error $5.6 \cdot 10^{-6}$, peak to $2 \cdot 10^{-9}$.

Design variation, reusing V built at θ_0 without rebuilding (controller entries perturbed by relative amounts ε; actuator/sensor positions moved by 0.15ε against Gaussian width 0.02):

perturbation	peak drift	max rel. error σ_1	peak error	lower bound holds
controller, $\varepsilon = 1.0$	68%	$5.6 \cdot 10^{-6}$	$6 \cdot 10^{-12}$	yes
positions, $\varepsilon = 0.5$	67%	$5.6 \cdot 10^{-6}$	$9 \cdot 10^{-7}$	yes
positions, $\varepsilon = 1.0$	37%	$1.1 \cdot 10^{-4}$	$1 \cdot 10^{-4}$	yes
both, $\varepsilon = 1.0$	53%	$1.1 \cdot 10^{-4}$	$1 \cdot 10^{-4}$	yes

The accuracy is essentially independent of how far the design moves; the certified lower bound never failed (worst shortfall 2%, a constant). *Plant-only subspace* (built with no knowledge of b, c or the controller, $\dim 30$) reused on four very different closed loops: $6 \cdot 10^{-3}$ in every case — the subspace is a property of the plant, and the two Woodbury columns buy three orders of magnitude.

σ_2 through the same subspace: $8.9 \cdot 10^{-6}$ (fixed design), $3 \cdot 10^{-5}$ at $\varepsilon = 1$.

Closed-loop eigenvalues (pencil projected on V augmented with the right/left eigenvectors of the six rightmost eigenvalues, $\dim 54$): the Ritz value nearest the true rightmost eigenvalue tracks it to 10^{-13} at θ_0, $5 \cdot 10^{-6}$ at $\varepsilon = 0.2$, $2 \cdot 10^{-5}$ at $\varepsilon = 0.5$ — an excellent *predictor* of where a watched eigenvalue went. But the oblique projection also produces *spurious* Ritz values far in the right half-plane (one at $2.6 + 7.6i$, corresponding to nothing): nothing variational protects a Petrov–Galerkin eigenvalue of a non-normal pencil. A full-model residual $\|(A_{\mathrm{cl}} - \lambda M_{\mathrm{cl}})v\|$ per Ritz pair, one matvec, filters them. **Predictor yes, certificate no.**

Real, unstable plant (GL realified to the state $(\mathrm{Re}\, q, \mathrm{Im}\, q)$ — real A, M with GL's spectrum and its conjugate — and shifted right until four conjugate pairs are unstable, $N = 798$): σ_1 on the axis is finite and symmetric; a

real subspace built from the real and imaginary parts of the anchor columns, with anchors on $\omega \geq 0$ only (five: $0.4, 0, 1.6, 4.0, 0.8$; real $\dim 60$), gives real projected matrices and $9.6 \cdot 10^{-6}$ on *both* halves of the axis, σ_2 to $1.5 \cdot 10^{-5}$; the four unstable pairs' eigenspaces lie inside V to a fraction 1.000 each without having been added, and the real reduced pencil reproduces them to four digits, in conjugate pairs, with nothing spurious in the right half-plane.

10.4 Unstable plants and real arithmetic

The cavity plant is unstable and real; neither changes the recipe. The anchors are solves on the imaginary axis, where $sM - A$ is nonsingular because the plant eigenvalues are off the axis; the resolvent near the axis is dominated by precisely the unstable modes, so the greedy anchors capture them on their own (measured above) — add their right and left eigenvectors explicitly anyway, they are known from the global stability analysis, cost nothing, and the left ones are needed for the eigenvalue predictor. Stability of the closed loop remains the certificate's statement (§7), not the reduced model's. Realness: the anchor factorisations stay complex (the shift is imaginary; a real $2n$ reformulation gains nothing), but from the columns on everything is real — split each column into real and imaginary parts, the subspace is closed under conjugation, the projected matrices are real, the reduced model is a real descriptor system with the real-rational $k(s)$, its eigenvalues come in conjugate pairs, and anchors and sweep are needed on $\omega \geq 0$ only. This composes with the realness savings of §7.3.

10.5 Optimise on the reduced model, enrich at the full one: Aliyev–Benner–Mengi–Voigt (2019), plus certification

The loop "run the optimisation on the reduced model, then consult the full model at the result and enrich there" is not ours. It is Algorithm 2.1 of N. Aliyev, P. Benner, E. Mengi, M. Voigt, "A subspace framework for H_∞-norm minimization", *SIAM J. Matrix Anal. Appl.* 41(2) (2020) 928–956 (arXiv:1905.04208, 2019), written for exactly our problem: minimise $\|H[\mu]\|_{H_\infty}$ over the parameters μ of a large descriptor system whose matrices depend affinely on μ through scalar functions, $E(\mu) = \sum f_j(\mu)E_j, A(\mu) = \sum g_j(\mu)A_j, B(\mu) = \sum h_j(\mu)B_j, C(\mu) = \sum k_j(\mu)C_j$ — which ours do: the controller entries through the rank-one coupling terms, the field coefficients through $b(\beta) = \Sigma\beta$. The algorithm:

1. $\mu^{(k)} \leftarrow \arg\min_\mu \|H^{V,W}[\mu]\|_{H_\infty}$ — the reduced norm minimised over the design (they use GRANSO, or the global optimiser `eigopt` when μ has few components);
2. $\omega^{(k)} \leftarrow \arg\max_\omega \sigma_1(H[\mu^{(k)}](i\omega))$ — one *full-model* norm evaluation at the reduced optimum;
3. $V \leftarrow V \oplus D(\mu^{(k)}, i\omega^{(k)})^{-1}B(\mu^{(k)})$, $W \leftarrow W \oplus D(\mu^{(k)}, i\omega^{(k)})^{-*}C(\mu^{(k)})^*$; repeat.

Their Lemma 2.1: at every iterate the reduced function Hermite-interpolates the full one *in both ω and μ* — value, σ_2, $\partial_\omega\sigma = 0$, and $\nabla_\mu\|H\|_{H_\infty} = \nabla_\mu\sigma^{V,W}$ (the reduced gradient with respect to the design equals the true gradient at the design where the snapshot was taken). Their Theorem 3.8: *superlinear* convergence of $\mu^{(k)}$ to a minimiser, under Assumption 3.1 — the supremum of $\sigma_1(\mu_*, \cdot)$ is attained at a *unique* ω_* where σ_1 is simple — and with μ_* interior (box constraints active at μ_* are handled by restricting to the inactive variables, their Remark 3.9; their real examples all have optima on the box boundary). In practice ≤ 10 outer iterations and "a few seconds for order 10^4". Stability is explicitly left open: "efficient stability checks would be needed for controller design".

What is ours. Two things, and both are consequences of the kind of optimum we meet. (i) Their Assumption 3.1 — a single active peak — fails at our optima, which are balances of several peaks (§1 of the optimisation note); at such a point the reduced function must interpolate the full one at *every* active peak, or the reduced optimum drifts along the crease. We therefore enrich at all certified near-active frequencies — the working set Ω — rather than at the single maximiser of line 2; their Algorithm 2.2, which adds nearby interpolation points to control second derivatives, points the same way. (ii) Line 2 trusts a full-model norm evaluation; ours is the certified sweep of §6, and the certificate does double duty: it decides whether the reduced level is the true level, and *where* to enrich — a refuted point, or a certified peak whose reduced value disagrees beyond tolerance, is the next anchor. Superlinear locally (theirs), globally certified (ours), and the reduced model is never trusted. Their per-design snapshot (line 3 at the current μ) is our per-design refresh of the Woodbury columns; for field parameters the affine dependence $b(\beta) = \Sigma\beta$ can be projected once, $W^{\mathsf{H}}\Sigma$ and $\Sigma^{\mathsf{H}}V$ ($\dim V \times n_\beta$), so that the reduced model carries the β-dependence of B exactly and the refresh only sharpens the resolvent images. Two practical gifts of their setting: on a reduced model an evaluation costs microseconds, so random multistart of the inner

problem is free, and for designs with few parameters (the position cases, $n_\theta = 2$–4) the reduced problem can be optimised *globally* (`eigopt`: piecewise-quadratic support functions, Breiman & Cutler 1993) before certification. What the certification cannot make cheaper is itself; the reduced model shortens its bisection and predicts the hot windows.

10.6 Eigenvalues and the margin on the reduced model: Mengi (2022, 2023)

Two further papers by the same group supply the mechanisms this section had left as "predictor only" and "certificate outside".

Watch-list tracking by interpolation. E. Mengi, "Large-scale estimation of dominant poles of a transfer function by an interpolatory framework", *SIAM J. Sci. Comput.* 44(4) (2022) A2412–A2438: compute the dominant poles of the *reduced* system (QZ on the small pencil), then expand V, W so that the full system is Hermite-interpolated *at those poles* — solves with $(A - \lambda E)$ at the complex shift λ equal to the pole estimate, with $q \geq 1$ derivative directions $(A - \lambda E)^{-1} E (A - \lambda E)^{-1} B$ — and repeat; convergence at least quadratic (his Theorem 4.3), termination and per-pole convergence by the full-model residual $\|(A - \lambda E) V v\|_\infty$ (his §5.2–5.3), one LU per pole per iteration, more reliable than SAMDP on benchmarks. For us this replaces the continuation gate's segment-by-segment tracking: the watch-list eigenvalues are followed through the reduced pencil as θ changes (the predictor of §10.3, $5 \cdot 10^{-6}$), each estimate is polished by one shift-invert solve at the Ritz value — his interpolation step — and the residual is the convergence certificate for that pole. The eigenvalue-identity problem that broke endpoint checks disappears, because the reduced pencil's poles vary continuously with θ; the spurious Ritz values of §10.3 are handled by the same step — interpolating at a non-pole adds information, and the value either converges to a true pole or fails the residual test. His dominance measure $\|Cv\| \, \|w^{\mathsf{H}} B\| / |\mathrm{Re}\, \lambda|$ becomes, for our full-field channel $B = C^{\mathsf{H}} = M^{1/2}$, $1/|\mathrm{Re}\, \lambda|$ times the mode's biorthogonality factor: the dominant poles are the near-axis, non-normal modes — the watch list and the peak generators at once, one subspace for both. His initialisation (§5.1) is "run the L_∞ subspace method crudely", i.e. our anchors. The winding count of §7 stays as the unconditional statement; this mechanism makes the per-step gate rigorous per pole and cheap.

The margin as a constraint on the reduced model. E. Mengi, "A subspace framework for L_∞ model reduction", arXiv:2309.08011 (2023), §6: the reduced problem is minimised subject to $\alpha(A_{\mathrm{red}}, E_{\mathrm{red}}) \leq \beta$, the spectral abscissa of the *reduced* pencil, imposed by a log-barrier $L_\mu = F - \mu \log(\beta - \alpha)$ — and he shows that the Hermite interpolation properties between full and reduced objective *extend to the barrier function*, gradient and Hessian included. For us that is the cheap form of the structural fix the optimisation note specifies (its to-do item 2): the explicit margin constraint $\alpha(\theta) \leq -\varepsilon$, evaluated and differentiated on the reduced closed-loop pencil inside the inner loop — eigenvalues by QZ on a $\dim V$ pencil, gradients from the reduced eigenvectors, residual filter applied — and certified afterwards by the winding count. The same paper's diagnosis of direct smooth optimisation of an L_∞ objective — "at best a linear rate, requiring too many evaluations of the costly norm" — is the v2 direct-mode record of the optimisation note in one sentence.

10.7 What projection does not replace: the frozen-preconditioner test

A related idea — reuse one factorisation as a *preconditioner* for GMRES at neighbouring frequencies — was tested first and rejected for the σ_1 oracle, for two measured reasons. The preconditioned operator $I + i\Delta\omega\, R(\omega_0) M$ has eigenvalues $1 + i\Delta\omega/(i\omega_0 - \lambda_j)$: on GL's dense branch there are 112 plant eigenvalues within $2r_0$ of a shift at $\omega = 8$, so the spectrum spreads instead of clustering (266 of 799 eigenvalues further than 0.1 from 1 at $\Delta\omega = r_0$), and non-normality widens the gap between $\|R(\omega_0)M\|$ and $1/r_0$ by a factor 18 at $\omega = 0$. GMRES then needs 10–30 iterations at tolerance 10^{-6}; and since the oracle needs $m \approx 10$–20 right-hand sides per frequency, each paying its own iterations, break-even is $k^* = 1 + C/m$ with C the LU-to-solve cost ratio — $k^* \approx 4$ in 2-D — and the idea loses by 3–7×. It *wins* where $m = 1$: the winding certificate's samples ($G = c^{\mathsf{H}} R b$, tolerance 10^{-3}), measured at 3–16 iterations for $\Delta\omega$ up to two band widths, against $k^* \approx 50$. Projection replaces the oracle; anchor-preconditioning serves the winding tail; the two are complementary, and both hang on the same handful of resident factorisations.

10.8 Cost and parallelism at cavity scale

For Taylor–Hood P2–P2–P1 on $2 \cdot 10^5$ triangles (7–$9 \cdot 10^5$ dof), a complex sparse LU with nested dissection should run to 50–100 nonzeros per row in the factors: *estimates*, to be measured, of 1–3 GB and 30–120 s per factorisation on one node. Fifty of those per inner iteration, times hundreds of iterations, times the case count,

is the cluster; with projection the accounting changes shape: (i) anchors, 5–10 factorisations *once per mesh*, design-independent, factors stored (~ 20 GB) — parallel *across* anchors, multithreaded within one node, never distributed across nodes for a 2-D factorisation; (ii) inner loop, *zero* factorisations — per design two solves per anchor (field cases only), per frequency $\sim \dim V$ sparse matvecs and a $\dim V \times \dim V$ SVD, all of Ω on one node; (iii) certification, $O(10^2)$ factorisations per outer round, embarrassingly parallel over shifts. The node estimate goes from "fifty nodes held for every case" to "one fat node for the optimisation and a handful borrowed for minutes per outer round". The open number is the subspace dimension in 2-D: GL needs 30–60; the cavity's richer near-axis spectrum may need several times more, which is still small but decides the estimate above, and it is the first thing to measure.

11 Cost accounting

quantity	primitives per evaluation	notes
$\sigma_1(\omega)$	1 LU + $O(10)$ solves	warm-started power iteration
$\nabla_\theta \sigma_1(\omega)$ (*all* params)	+2 solve sets	independent of NZ
$\partial_\omega \sigma_1$, Newton peak polish	inner products + $O(3)$ values	per peak
certified band maximum	$O(10^2)$ shift-invert solves	rtol 10^{-6}, adaptive disks
α tracking (per accepted step)	few bordered solves per eigenvalue	shifted at previous location
winding certificate (per saved design)	500–900 LUs of $R(s)^{-1}$	unconditional count (9)
reduced sweep on Ω [proposed, §10]	0 LU; $\sim \dim V$ matvecs + one $\dim V$ SVD per frequency	anchors: 5–10 LUs once per
$\nabla_\theta \alpha$	one eigenpair (+ adjoint)	formula (10)

Every line is (P1)–(P3). The two design decisions that make the whole pipeline scale are visible in this table: gradients whose cost does not grow with the number of parameters (§4), and an outer method (the exchange) shaped so that the LU-bearing lines are executed for many independent frequencies concurrently. The controller only ever appears through the scalar $k(s)$ of (2) and the rank-2 identity (3) — which is also, through (7), the reason the stability of an $(n+r)$-dimensional loop can be certified by winding a single complex number around a line.

Parallel cost, corrected. The winding line of the table is a *serial* count. Split at the knots (§7.2), the same work is one round of ~ 120 knot factorisations followed by a work queue whose critical path is the longest interval — a few tens of serial factorisations — so the wall time at cavity scale is set by the hottest pole window, not by the total.

12 Decisions and pending items

The review rounds produced decisions and open items; they are collected here, each pointing at the section that justifies it. Items 1–2 amend statements of earlier editions; the rest are code.

1. **Tail bound for a singular mass matrix** (§7.2). The bound $\|R(s)\| \le 1/(\lambda_{\min}(M)(|s| - \nu))$ presumes M invertible; the cavity's $M = \mathrm{diag}(M_\nu, 0)$ is not. Re-derive on the discretely divergence-free velocity subspace with the Leray projector Π: $\nu := \|\Pi M_\nu^{-1} A_{\nu\nu} \Pi\|$, $\lambda_{\min}(M_\nu)$, and a strictly proper G guaranteed by momentum-equation inputs and velocity outputs. Record the two modelling rules that fall out (pressure sensors make G proper, not strictly proper, and shift the tail disk to $1 - KG(\infty)$; inputs must never enter the constraint equation). Validate on a Stokes or lid-driven benchmark where a dense generalised eigensolve still exists.
2. **Once-per-mesh constants, tight values with certified bounds**. Replace the ARPACK `which='SM'` call in `plant_tail_constants` by shift-invert Lanczos (Cholesky- or CG-based) for a *tight* $\lambda_{\min}(M)$, and compute $\|A\|_2$ by Lanczos on $A^{\mathsf{H}}A$ (matvecs only); alongside, compute the element-wise floor $\lambda_{\min}(M) \ge \min_e \lambda_{\min}(M_e)$ and the Hölder ceiling as certified cross-checks. Tight constants shorten the tracked segment on every design; the bounds catch bugs in the tight computations and vice versa.
3. P_{plant} **at cavity scale** (§7.4). Implement the three steps as code, not prose: numerical-range rectangle with the inertia certificate (one Hermitian Lanczos plus one LDL^{H} per bound); Beyn contour count with probes, cross-validated at two quadrature orders and two probe blocks, left edge at $\ell - \delta$; shift-invert polish of

every counted eigenvalue. Engine decision: **Beyn**, per §7.4. Prototype at GL against the dense count that `plant_pole_data` already provides.

4. **Parallel winding certificate** (§7.2). Interval decomposition at the knots, two-pass (knot values, then a work queue with midpoint splitting of intervals that bisect below a threshold), phase subtotals summed. `winding_cert.py` is serial as frozen; measure the critical path at GL before projecting cavity wall times.

5. **A claim in the code that the code does not honour — resolved: implement, merged into item 4.** The docstring of `winding_cert.py` advertises a `-rigor` mode (a Neumann segment bound per sample, "~ 8 extra matvecs per point"); the argument parser has no such option and no such bound is implemented. What was meant is a mechanism of this note — the per-interval Neumann bound on $|kG|$ that closes quiet intervals without further samples, i.e. exactly the "coarse-certify" pass-2 test of the parallel scheme (item 4) and thrift item (4) of the cavity-cost answer. So: implement it as part of item 4's pass 2, where it belongs; the docstring then becomes true. Recorded in the spirit of the report's §7.

6. **Cost table** (§10). Add a column for the parallel critical path once item 4 is measured, so the table stops quoting serial counts for a quantity that will never run serially at scale.

7. **MIMO capability** (§1). Generalise the coupling to n_a actuators and n_s sensors: Woodbury in place of Sherman–Morrison, the return-difference determinant $f = \det(I - \mathsf{KG})$ in place of the scalar, n_a+n_s solves per factorisation in place of two. Structure already in place; cost grows only in the solve count.

8. **Projection oracle, GL prototype** (§10). Implement the reduced-basis variant in the sparse pipeline: greedy anchors on $\omega \geq 0$, real subspace, top-3 singular directions plus the two Woodbury columns per anchor, unstable eigenvectors added; values, σ_2, gradients and the eigenvalue predictor from the reduced model; certification on the full model with refuted points and certified-peak disagreements as the enrichment rule — Algorithm 2.1 of Aliyev–Benner–Mengi–Voigt with enrichment at Ω instead of the single maximiser and the certified sweep in place of line 2 (§10.5); project the field basis once ($W^{\mathsf{H}}\Sigma$, $\Sigma^{\mathsf{H}}V$) so that $b(\beta)$ is exact in the reduced model. Validate against the 27 frozen cases: same certified optima, factorisation count per inner iteration reduced from $|\Omega|$ to zero. Then the *trajectory* test that has not been run: does V ever need refreshing along a full optimisation, and does the certified-vs-reduced gap at the peaks detect it.

9. **Subspace dimension in 2-D — measure first.** On the cavity mesh (or a coarser cavity), build the plant subspace once and record the greedy dimension needed for 10^{-5} over the band; every cost estimate of §10.8 depends on it. Store the anchor factorisations and measure their size and time (estimates: 1–3 GB, 30–120 s).

10. **Anchor-preconditioned winding tail** (§10.7). GMRES from the nearest resident anchor for the single-solve samples of the segment, tolerance 10^{-3}, with the frozen $|f|$ used to tighten the tolerance near closed-loop poles; the σ_1 oracle and the certified disk radii explicitly excluded until the inexact-shift-invert bound is written. Merge with item 4's two-pass scheme.

11. **Watch-list tracking by interpolation** (§10.6). Replace the continuation gate's tracking by Mengi's dominant-pole framework: reduced-pencil poles as θ moves, one shift-invert polish at each Ritz value, full-model residual as the per-pole convergence test; spurious Ritz values eliminated by the same step. Validate by replaying the 27 frozen accepted-step chains against the gate's decisions.

12. **Margin constraint on the reduced model** (§10.6). Impose $\alpha_{\mathrm{red}}(\theta) \leq -\varepsilon$ inside the inner loop — QZ on the reduced closed-loop pencil, gradients from reduced eigenvectors, residual filter — as the cheap form of the optimisation note's item 2; winding count outside. Test on K+B+C, the case that stalled at the boundary.

13. **Global optimisation of the reduced problem for the position cases** (§10.5). With $n_\theta = 2$–4, run a global optimiser (`eigopt`-style) on the reduced model, then certify; and use free multistart of the reduced inner problem in the random-start study of the optimisation note's item 1.